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NAAC Accredited Autonomous Institution
Approved by AICTE & Affiliated to Anna University
ISO 9001:2015 Certified Institution
Thalavapalayam, Karur - 639 113, TAMILNADU.

EEB1332 DESIGN PROJECT

AUTOMATED AMMONIA DETECTION AND NEUTRALIZING SYSTEM IN POULTRY FARMS

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONICS ENGINEERING

M.KUMARASAMY COLLEGE OF ENGINEERING, KARUR

MAY 2026

M.KUMARASAMY COLLEGE OF ENGINEERING, KARUR

(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report "AUTOMATED AMMONIA DETECTION AND NEUTRALIZING SYSTEM IN POULTRY FARMS" is the bonafide work of SUMANRAJ D (927623BEE108) , YAASHIKA K (927623BEE121), YOGAPRIYA B (927623BEE123) who carried out the project work during the academic year 2025-2026 under my supervision.

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This Project Work (EEB1332 – Design Project) report has been submitted for the Sixth Semester Project viva voce Examination held on

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INTERNAL EXAMINER **EXTERNAL EXAMINE**

VISION AND MISSION OF THE INSTITUTION

VISION

To emerge as a leader among the top institutions in the field of technical education

MISSION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges
- Create a diverse, fully-engaged learner-centric campus environment to provide quality education to the students
- Maintain mutually beneficial partnerships with our alumni, industry and professional associations.

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

VISION

To produce smart and dynamic professionals with profound theoretical and practical knowledge comparable with the best in the field.

MISSION

- Produce hi-tech professionals in the field of Electrical and Electronics Engineering by inculcating core knowledge.
- Produce highly competent professionals with thrust on research.
- Provide personalized training to the students for enriching their skills.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Graduates will have flourishing career in the core areas of Electrical Engineering and allied disciplines.

PEO2: Graduates will pursue higher studies and succeed in academic/research careers.

PEO3: Graduates will be a successful entrepreneur in creating jobs related to Electrical and Electronics Engineering /allied disciplines.

PEO4: Graduates will practice ethics and have habit of continuous learning for their success in the chosen career.

PROGRAMME OUTCOMES (POs)

After the successful completion of the B.E. Electrical and Electronics Engineering degree Program, the students will be able to:

- **PO1 Engineering knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop solutions to complex engineering problems.
- **PO2 Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- **PO3 Design/development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

- **PO4 Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
- **PO5 Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
- **PO6 The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- **PO7 Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
- **PO8 Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9 Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10 Project management and finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11 Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

The following are the Program Specific Outcomes of Engineering Students:

PSO 01: Power Systems Engineering

Apply comprehensive knowledge of electrical power systems to analyze, design and manage generation, transmission, and distribution networks, integrating emerging technologies and sustainable practices to contribute effectively to industry and societal needs.

PSO 02: Power Electronics and Drives

Design and develop efficient, reliable power electronic systems and motor drives with an emphasis on sustainable energy use, and inclusive technology solutions that address societal and environmental challenges.

PSO 03: Electronics and Instrumentation Engineering

Develop intelligent instrumentation and embedded systems for accurate measurements, monitoring and control in interdisciplinary domains addressing societal needs.

ABSTRACT	**POs MAPPING**
Poultry Farming , Ammonia Detection , Air Quality Monitoring , Electrochemical Sensor , STM Microcontroller , Gas Neutralization , Smart Farming , IoT Monitoring , Environmental Control , Poultry Health , Precision Agriculture , Automated System	PO1, PO2, PO3, PO4, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3

SDG MAPPING

SDG No	SDG GOAL	MAPPING JUSTIFICATON
SDG 3	Good Health and Well-being	The project improves air quality in poultry farms by reducing ammonia levels, which helps prevent respiratory diseases in birds. A healthier environment also reduces the risk of harmful exposure to farm workers, promoting overall well-being.
SDG 2	Zero Hunger	By improving poultry health and reducing mortality rates, the system increases egg and meat production. This supports food security and ensures a stable supply of protein-rich food.
SDG 9	Industry , Innovation and Infrastructure	This project introduces an innovative, technology-driven solution using sensors and automation, contributing to the development of smart agricultural infrastructure.

PROFESSIONAL BODY STANDARDS

Professional Body Standards Number	Professional Body Standards Name	Remarks
IEEE 1451	Smart Transducer Interface for Sensors and Actuators	Used for proper integration of ammonia sensor with microcontroller for accurate data acquisition
IEEE 7000	Ethical Considerations in System Design	Ensures the system is designed safely, ethically, and benefits both poultry and farmers

ABSTRACT

Poultry farming requires a controlled environment to ensure the health and productivity of birds. One of the major challenges in poultry farms is the accumulation of ammonia gas produced from the decomposition of waste materials. High concentrations of ammonia can lead to respiratory disorders, reduced growth rate, decreased egg production, and increased mortality, ultimately causing economic losses to farmers. Existing systems mainly rely on manual monitoring and ventilation, which are inefficient and lack real-time response.

This project proposes an automated ammonia detection and neutralization system designed to monitor and control air quality in poultry farms. The system utilizes an electrochemical ammonia sensor for accurate detection of gas concentration and an STM-based microcontroller for real-time data processing and control. When the ammonia level exceeds a predefined threshold, the system activates a spray mechanism that disperses a neutralizing solution, such as citric acid mist, to reduce ammonia concentration effectively. Additionally, the system provides visual and audio alerts and can be integrated with IoT platforms for remote monitoring.

The proposed solution is designed to be cost-effective, scalable, and suitable for both small-scale and large-scale poultry farms. By maintaining optimal air quality, the system improves bird health, enhances productivity, reduces mortality rates, and increases overall farm profitability. This project demonstrates a practical approach toward smart farming and contributes to the advancement of precision agriculture technologies.

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LIST OF ABBREVIATIONS

S.No	Abbreviation	Expansion
1	AC	Alternating Current
2	DC	Direct Current
3	ESP32	Espressif 32-bit Microcontroller
4	LCD	Liquid Crystal Display
5	DHT	Digital Humidity and Temperature
6	MQ135	Gas Sensor Model 135 (Ammonia Sensor)
7	IoT	Internet of Things
8	NH_3	Ammonia Gas
9	PPM	Parts Per Million
10	MCU	Microcontroller Unit
11	VCC	Supply Voltage
12	GND	Ground
13	GPIO	General Purpose Input Output
14	ADC	Analog to Digital Converter
15	PWM	Pulse Width Modulation
16	LED	Light Emitting Diode
17	H_2S	Hydrogen Sulfide
18	CO_2	Carbon Dioxide
19	SDG	Sustainable Development Goals
20	IEEE	Institute of Electrical and Electronics Engineers

CHAPTER 1
INTRODUCTION

Maintaining a healthy environment in poultry houses is essential for bird welfare and productivity. Ammonia gas from droppings can cause respiratory problems, stress, and reduced immunity in birds. This project implements a smart ammonia detection and neutralization system using gas sensors and automated dispensers. An IoT dashboard provides real-time monitoring and alerts, ensuring better hygiene, improved bird health, and efficient farm management.

1.1 Objective Overview

- To monitor ammonia levels in poultry farms in real time.
- To develop an automated detection system using gas sensors.
- To design a control system using STM microcontroller.
- To reduce harmful ammonia concentration using a neutralizing spray.
- To maintain a healthy environment for poultry birds.
- To minimize respiratory diseases and mortality rate.
- To improve growth rate and egg production.
- To provide alert system (buzzer/LED/display) for safety.
- To ensure the system is cost-effective and scalable.
- To support smart and sustainable poultry farming.

1.2 Scope of the Project

The scope of this project focuses on monitoring and controlling ammonia levels in poultry farms to maintain a healthy environment for birds. The system provides real-time detection using gas sensors and automatically activates a neutralization spray mechanism when ammonia exceeds safe limits. It can be implemented in small, medium, and large-scale poultry farms and can be further enhanced with IoT integration for remote monitoring and data analysis. Additionally, the system can be extended to detect other harmful gases such as hydrogen sulfide and carbon dioxide. Overall, the project has strong potential for development into a commercial smart farming solution that improves air quality, bird health, and farm productivity.

1.3 Advantages of Proposed System

- Provides real-time monitoring of ammonia levels
- Automatically neutralizes harmful gases
- Improves air quality inside poultry farms
- Reduces respiratory diseases in birds
- Decreases mortality rate
- Enhances growth and egg production
- Reduces manual effort and human error
- Gives early warning through alerts (buzzer/LED/display)
- Cost-effective compared to industrial systems
- Scalable for small and large farms
- Supports smart farming and automation
- Can be extended to detect other harmful gases

CHAPTER 2
LITERATURE REVIEW

1. Paper Title: Ammonia Emission in Poultry Facilities: A Review for Tropical Climate Areas

Authors: M.D. Oliveira et al.

Year: 2021

M.D. Oliveira et al. (2021) presents a comprehensive review on ammonia emissions in poultry facilities, highlighting that ammonia is mainly generated from the decomposition of poultry litter. The study emphasizes that continuous monitoring of ammonia concentration is essential for maintaining air quality. It also discusses various sensing techniques such as electrochemical and semiconductor sensors for ammonia detection. The research concludes that accurate monitoring systems are necessary for reducing emissions and improving environmental conditions in poultry farms.

2. Paper Title: Ammonia and Poultry Welfare: A Review

Authors: H.H. Kristensen et al.

Year: 2019

H.H. Kristensen et al. (2019) examines the impact of ammonia on poultry welfare and reports that ammonia exposure causes irritation to the respiratory system and eyes of birds. The study highlights that high ammonia levels reduce growth rate and feed efficiency. It stresses the importance of maintaining ammonia below safe limits to ensure bird health and productivity.

3. Paper Title: Ammonia Emissions, Impacts, and Mitigation Strategies for Poultry Production

Authors: R.B. Bist et al.

Year: 2023

R.B. Bist et al. (2023) discusses ammonia as a major pollutant in poultry farms and its environmental and health impacts. The study suggests that ammonia levels should be maintained below 25 ppm to prevent health issues. It also reviews various mitigation strategies including ventilation and chemical treatments. The research highlights the need for automated systems for effective monitoring and control.

4. Paper Title: Ammonia and Ethanol Detection Using Electronic Nose System

Authors: Y. Shi et al.

Year: 2024

Y. Shi et al. (2024) proposes an electronic nose system for detecting ammonia using optimized neural network algorithms. The study emphasizes the importance of fast and accurate detection of ammonia in livestock environments. The system improves detection speed and accuracy, making it suitable for real-time monitoring applications in poultry farms.

5. Paper Title: Detection of NH_3 in Poultry Housing Using Laser Spectroscopy

Authors: K.W. Wang et al.

Year: 2022

K.W. Wang et al. (2022) developed a high-precision ammonia detection system using tunable diode laser absorption spectroscopy. The system provides high accuracy (0.2 ppm) and fast response time. The study highlights the importance of precise monitoring for maintaining environmental conditions in poultry houses, although the cost and complexity are higher.

6. Paper Title: Quick Tests for Determination of Ammonia in Poultry Litter

Authors: J.B. Reeves et al.

Year: 2002

J.B. Reeves et al. (2002) evaluates different quick testing methods for measuring ammonia in poultry litter. The study finds that while several techniques exist, their accuracy varies, and some methods are not reliable for real-time monitoring. This highlights the need for improved sensing systems for accurate ammonia detection.

7. Paper Title: Ammonia Generation System for Poultry Health Research Using Arduino

Authors: M. Calvet et al.

Year: 2021

M. Calvet et al. (2021) presents a system to study ammonia effects on poultry using an Arduino-based setup. The study shows that ammonia levels above 50–75 ppm significantly reduce poultry productivity and increase disease risk. It emphasizes the need for controlled environments and monitoring systems in poultry farms.

8. Paper Title: Development of MOS Sensor-Based NH_3 Monitor for Poultry Houses

Authors: X. Liu et al.

Year: 2016

X. Liu et al. (2016) developed a metal oxide semiconductor (MOS)-based ammonia monitoring system for poultry farms. The study achieved good accuracy with less than 10% error and demonstrated that low-cost sensors can be effectively used with proper calibration. The system is portable and suitable for field applications.

9. Paper Title: Ammonia Control in Broiler Houses

Authors: F.N. Reece et al.

Year: 1979

F.N. Reece et al. (1979) investigates methods to control ammonia release in poultry houses using chemical treatments. The study found that reducing litter pH helps in controlling ammonia emission. It highlights early approaches to ammonia control, forming the basis for modern neutralization techniques.

10. Paper Title: Ammonia Emissions in Poultry Houses and Reduction Strategies
Authors: Z. Chen et al.
Year: 2021

Z. Chen et al. (2021) reviews ammonia emission mechanisms and reduction strategies in poultry farms. The study explains that ammonia levels above 25 ppm negatively impact bird health and productivity. It suggests biological and chemical methods for ammonia reduction and emphasizes sustainable approaches for long-term control.

CHAPTER 3
EXISTING SYSTEM

3.1 INTRODUCTION

In traditional poultry farming, maintaining proper air quality is a major challenge due to the accumulation of ammonia gas produced from the decomposition of poultry waste. The existing systems mainly rely on natural ventilation and manual monitoring methods to control ammonia levels. Farmers typically use exhaust fans, open ventilation structures, or chemical additives in litter to reduce gas concentration. However, these methods are not efficient as they do not provide real-time monitoring or automatic control. There is no continuous tracking of ammonia levels, and actions are usually taken only after the problem becomes severe. As a result, birds are often exposed to harmful gas levels, leading to health issues, reduced productivity, and economic losses. Therefore, the existing systems lack automation, accuracy, and timely response, highlighting the need for an improved solution.

3.2 BLOCK DIAGRAM

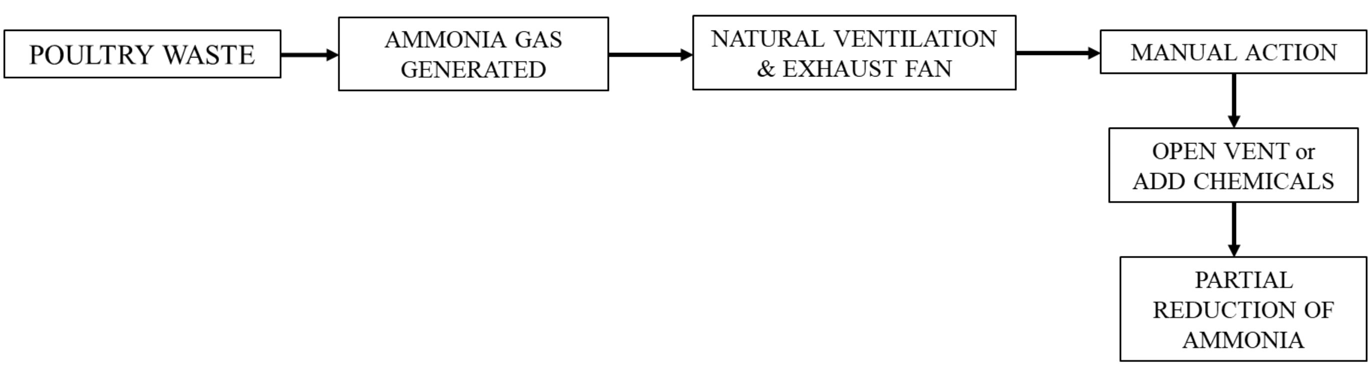

Fig 3.1 BLOCK DIAGRAM OF EXISTING SYSTEM

3.3 WORKING

1. Ammonia Detection

The system continuously monitors the poultry environment using an ammonia gas sensor. The sensor detects the concentration of ammonia present in the air and converts it into an electrical signal.

2. Data Processing

The detected signal is sent to the microcontroller (STM/ESP32), where it is processed and compared with a predefined safe threshold level.

3. Condition Monitoring

If the ammonia level is within the safe limit, the system remains in normal mode and continues monitoring without any action.

4. Automatic Control Activation

When the ammonia concentration exceeds the threshold value, the microcontroller activates the control system by turning ON the relay.

5. Neutralization Process

The relay activates the pump, which sprays a fine mist of neutralizing solution (such as citric acid) through nozzles into the air. This helps in reducing ammonia concentration.

6. Alert Mechanism

At the same time, the system provides alerts using a buzzer or LED and displays the ammonia level/status on the display unit.

7. Restoration to Safe Level

As the ammonia level decreases and returns to a safe range, the system automatically turns OFF the pump and stops spraying.

8. Continuous Monitoring

The system continues to monitor the environment in real time, ensuring consistent air quality inside the poultry farm.

3.4 Problem Statement

Ammonia accumulation in poultry houses, produced from droppings and poor ventilation, poses serious risks to bird health and farm productivity. High ammonia levels can cause respiratory problems, stress, reduced immunity, and lower growth or egg production. Traditional methods of controlling ammonia, such as manual cleaning and ventilation, are inefficient, labor-intensive, and often unable to maintain safe air quality. There is a need for a smart, automated system that can detect ammonia levels in real-time and neutralize excess gas to ensure a healthy environment, improved bird welfare, and efficient farm management.

CHAPTER 4
PROPOSED SYSTEM

4.1 INTRODUCTION

The Automated Ammonia Detection and Neutralizing System in Poultry Farms is designed to monitor and control harmful ammonia gas present inside poultry sheds. Ammonia is mainly produced from poultry waste, and a high concentration of this gas can affect the health of birds as well as farm workers

4.2 BLOCK DIAGRAM

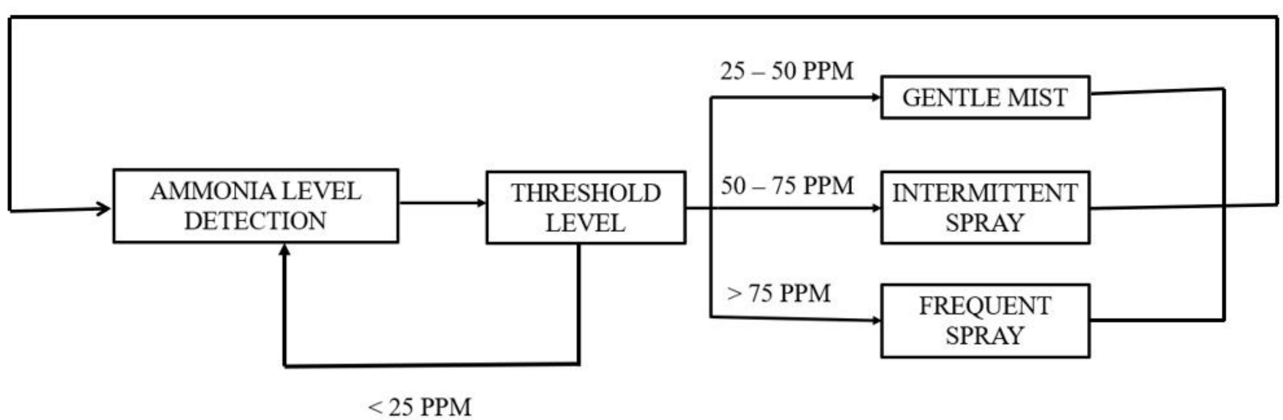

Fig. 4.1 BLOCK DIAGRAM
OF PROPOSED STEM

4.3 BLOCK DIAGRAM DESCRIPTION

The Automated Ammonia Detection and Neutralizing System in Poultry Farms explains the step-by-step operation of the system used to monitor and control harmful ammonia gas in the poultry shed. Ammonia is mainly produced from poultry waste, litter decomposition, and poor ventilation, and when its concentration increases, it can affect the health of birds and workers. In this system, the ammonia level detection block continuously senses the ammonia concentration present in the air. The detected value is then sent to the threshold level block, where it is compared with predefined ammonia ranges. This comparison helps the system decide whether the environment is safe or whether any neutralizing action is required. Thus, the first part of the block diagram focuses on gas sensing and decision making based on the measured ammonia level.

The second part of the block diagram shows the action taken for different ammonia concentrations. If the ammonia level is less than 25 PPM, the system considers it safe and continues only monitoring through the feedback loop without activating any mist or spray system. When the ammonia level rises to 25–50 PPM, the system activates a gentle mist, which provides a light neutralizing effect to reduce the gas concentration. If the ammonia level further increases to 50–75 PPM, the system turns on an intermittent spray, which gives a stronger response by spraying at intervals. When the ammonia level becomes greater than 75 PPM, the system activates a frequent spray, which works more aggressively to reduce the harmful gas level quickly. In this way, the system provides different levels of response based on the severity of ammonia concentration.

Instead, it applies only the required level of control, which saves water, energy, and operating cost. Overall, this block diagram clearly shows that the proposed system is an automatic, intelligent, and effective solution for poultry farms, as it continuously detects ammonia, compares it with threshold levels, and activates the proper neutralizing action to maintain a safe and healthy farm environment.

4.4 CIRCUIT DIAGRAM

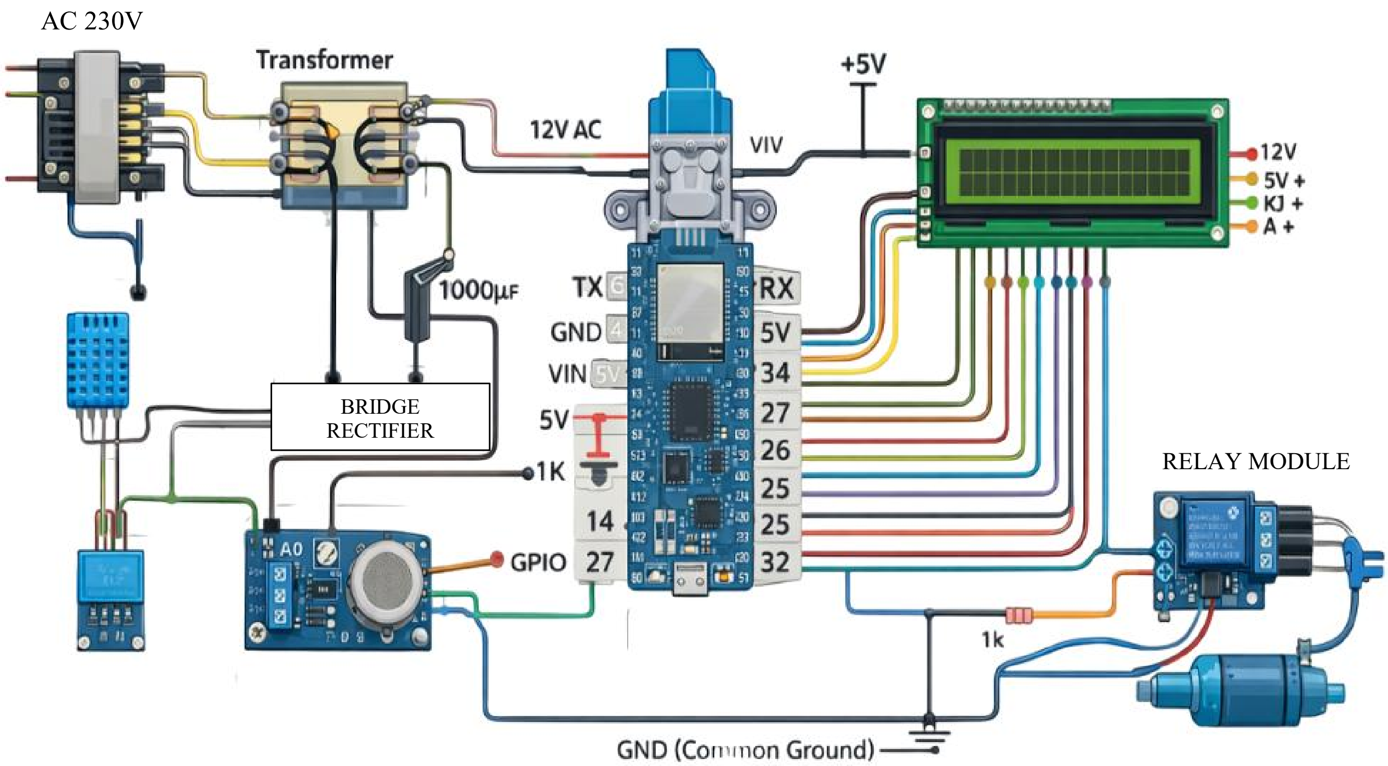

Fig. 4.2 CIRCUIT DIAGRAM

4.5 WORKING

The automated ammonia detection and neutralizing system in poultry farms is designed to continuously monitor air quality and automatically control harmful ammonia gas levels to maintain a safe environment for poultry and farm workers. In this system, an ammonia sensor such as MQ-135 or MQ-137 is strategically placed inside the poultry shed, where it constantly detects the concentration of ammonia gas produced from poultry waste. The sensor converts the detected gas concentration into an electrical signal, which is then transmitted to a microcontroller like Arduino or Raspberry Pi for processing. The microcontroller analyzes this data by converting it into parts per million (ppm) and comparing it with a predefined safe threshold level, typically around 20–25 ppm. As long as the ammonia concentration remains below this limit, the system continues normal monitoring without initiating any action.

However, when the ammonia level exceeds the safe threshold, the microcontroller automatically triggers a series of control mechanisms to reduce the gas concentration. Firstly, it activates ventilation systems such as exhaust fans to remove ammonia-contaminated air and allow fresh air to circulate within the poultry house. Simultaneously, a neutralization system is activated, which may include a misting or spraying unit that releases water or a mild acidic solution like citric acid; this helps in chemically reacting with ammonia and reducing its concentration in the air. In addition to these actions, an alert system such as a buzzer, LED indicator, or even SMS/mobile notification may be activated to inform the farmer about the high ammonia levels. The system operates on a real-time feedback mechanism, meaning the sensor continuously monitors the ammonia concentration even during the neutralization process. Once the ammonia level drops back below the safe limit, the microcontroller automatically switches off the ventilation and spraying systems, thereby conserving energy and resources. This entire process runs continuously in a loop, ensuring 24/7 monitoring and control without the need for manual intervention. As a result, the system helps in improving poultry health, reducing respiratory problems, enhancing productivity, and maintaining an overall hygienic and safe farm environment in an efficient and automated manner.

CHAPTER 5

HARDWARE IMPLEMENTATION

5.1 HARDWARE EXPERIMENTAL SETUP

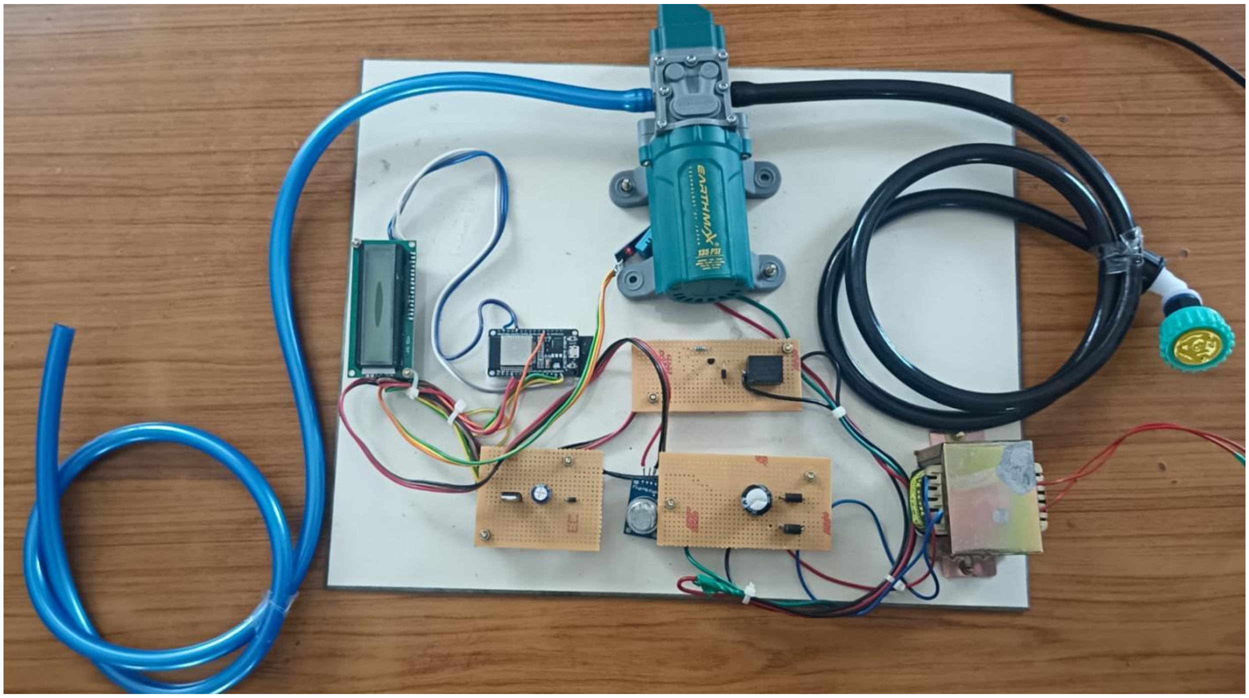

Fig 5.1 EXPERIMENTAL SETUP

5.2 COMPONENTS DESCRIPTION

1. ESP32 MODULE

The ESP32 module, developed by Esp32ASFDSystems, is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities. It plays a key role in the automated ammonia detection system by receiving data from sensors like MQ-135 or MQ-137. The ESP32 processes the sensor data and compares ammonia levels with a preset safe limit.

2. 16x2 LCD Display

The 16x2 LCD display is a commonly used output device in embedded systems that can display 16 characters in each of its 2 rows. It is used in the ammonia detection system to show real-time data such as ammonia concentration levels, system status, and alerts. The display is connected to the microcontroller (like ESP32 or Arduino) and receives data signals to present readable information. When ammonia levels exceed the safe limit, warning messages are shown on the screen. This helps farmers easily monitor conditions without needing additional devices.

3. DHT11 SENSOR

The DHT11 sensor is a low-cost digital sensor used to measure temperature and humidity in the environment. In the ammonia detection system, it helps monitor environmental conditions inside the poultry farm. The sensor sends real-time temperature and humidity data to the microcontroller (such as ESP32 or Arduino). This information is important because high humidity and temperature can increase ammonia formation.

4. TRANSFORMER

A transformer is an electrical device used to step down high voltage AC supply to a lower voltage suitable for electronic components. In the ammonia detection system, it converts the mains power (230V AC) into a safer low voltage (like 12V or 5V). This low voltage is then used to power the microcontroller, sensors, and other circuit components. It ensures safe and stable operation of the system. Transformers also provide electrical isolation, protecting the circuit from damage

5. DC PUMP

A DC pump is a small electric pump powered by direct current (DC) used to circulate or spray liquids. In the ammonia detection and neutralizing system, it is used to spray water or mild acidic solutions to neutralize excess ammonia in the poultry shed. The pump is controlled by the microcontroller (like ESP32), which turns it on automatically when ammonia levels exceed the safe limit. This ensures precise and timely neutralization without manual intervention. Using a DC pump makes the system energy-efficient and reliable for continuous operation.

6. MQ135 AMMONIA SENSOR

The MQ-135 sensor is a gas sensor used to detect ammonia (NH_3) and other harmful gases in the air. In the automated ammonia detection system, it continuously measures the concentration of ammonia inside the poultry farm. The sensor produces an analog signal proportional to the gas concentration, which is sent to a microcontroller like ESP32 or Arduino for processing**.**

7. VOLTAGE REGULATOR

A voltage regulator is a crucial electronic component designed to automatically maintain a constant output voltage regardless of fluctuations in the input voltage or changes in the load current. Its primary usage is to protect sensitive electronic circuits from damaging voltage spikes or drops, and to ensure a stable, reliable power supply. For example, in an Arduino project powered by a battery, a voltage regulator (like the IC pictured) will take the unstable input and regulate it down to a steady, safe output.

5.3 HARDWARE COMPONENTS USED

S. No	Components Used	Quantity
1.	ESP32 Module	1
2.	16x2 LCD Display	1
3.	DHT11 SENSOR	1
4.	TRANSFORMER	1
5.	DC PUMP	1
6.	MQ135 AMMONIA SENSOR	1
7.	VOLTAGE REGULATOR	1

Table 5.1 Hardware Components

CHAPTER 6
RESULTS AND DISCUSSION

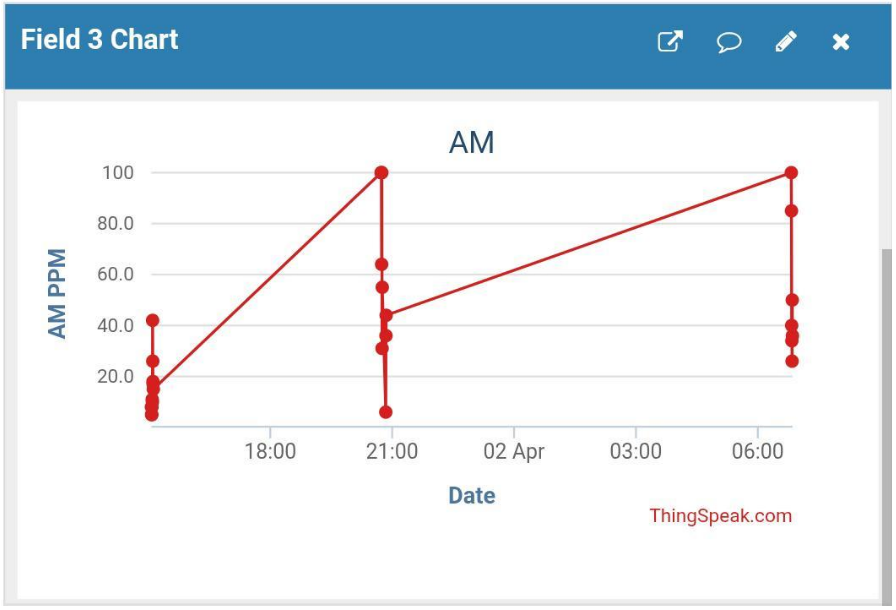

Fig 6.1 Ammonia Level View On Mobile Dashboard

The given chart represents the variation of ammonia (NH_3) concentration levels, measured in parts per million (PPM), over different time intervals in a poultry farm using an IoT platform like ThingSpeak. The X-axis indicates the date and time of measurement, while the Y-axis shows the ammonia concentration. At around 18:00, the ammonia levels are relatively low, ranging between 10 to 40 PPM, which is considered a safe or acceptable range for poultry health. However, at 21:00, there is a sudden increase in ammonia concentration, reaching values as high as 100 PPM, with several readings between 30 and 70 PPM, indicating poor ventilation and accumulation of harmful gases. Similarly, at 06:00 the next day, the ammonia levels again rise significantly, showing repeated peaks close to 100 PPM, which is dangerous for birds and can cause respiratory problems and stress. Ideally, ammonia levels should be maintained below 25 PPM, while levels above 50 PPM are harmful.

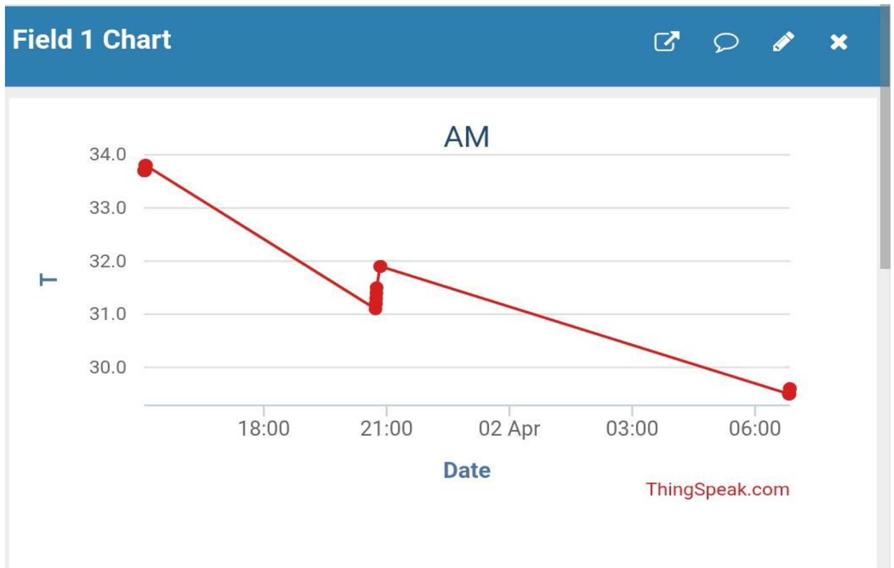

Fig 6.2 Temperature Level View On Mobile Dashboard

The given chart represents the variation of temperature (T) over time in the poultry farm monitoring system. The X-axis shows the date and time of observation, while the Y-axis indicates the temperature values. Initially, around 18:00, the temperature is relatively high at approximately 34°C. As time progresses to 21:00, the temperature decreases slightly and stabilizes between 31°C and 32°C, indicating a controlled environment during that period. By 06:00 the next day, the temperature further drops to around 29–30°C, showing a gradual cooling trend overnight. This variation suggests that the environmental conditions are being monitored and possibly regulated to maintain a suitable temperature range for poultry health. Maintaining an optimal temperature is important, as excessive heat can cause stress to birds, while lower temperatures can affect their growth and productivity. Thus, continuous monitoring helps ensure a balanced and healthy environment inside the poultry farm.

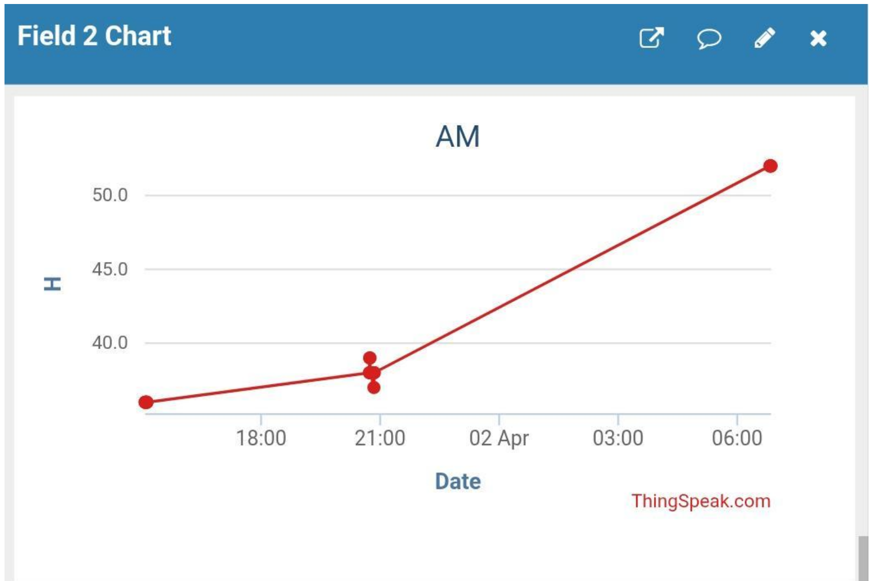

Fig 6.3 : Humidity level View On Mobile Dashboard

The given chart represents the variation of humidity (H) levels over time in the poultry farm monitoring system. The X-axis shows the date and time, while the Y-axis indicates the humidity percentage. At around 18:00, the humidity level is approximately 36%, indicating a relatively moderate condition. By 21:00, the humidity slightly increases and fluctuates between 37% and 39%, showing minor variations within a controlled range. However, by 06:00 the next day, the humidity rises significantly to about 52%, indicating a noticeable increase in moisture content in the environment. This upward trend suggests that humidity tends to accumulate overnight, possibly due to reduced ventilation or increased moisture from litter and bird activity. Maintaining proper humidity levels is important in poultry farms, as high humidity can lead to poor air quality, increased ammonia production, and health issues in birds. Therefore, continuous monitoring and control of humidity are essential to ensure a healthy and productive poultry environment.

Fig 6.4 Ammonia Level Indication

The display shows real-time environmental parameters measured in the poultry farm using the automated monitoring system. It indicates a temperature of 32.6°C, humidity of 37%, and ammonia level of 31 PPM. These values suggest that while temperature and humidity are within a moderate range, the ammonia level is slightly above the safe limit (25 PPM), indicating the need for ventilation or neutralization. This helps in maintaining a healthy environment for poultry through continuous monitoring.

Fig 6.5 Ammonia Level Normal

The display shows real-time monitoring of environmental conditions in the poultry farm, with a temperature of 32.5°C and humidity of 38%. The message “AMMONIA NORMAL” indicates that the ammonia level is within the safe limit. This confirms that the system is maintaining a healthy environment, ensuring proper air quality and suitable conditions for poultry.

Fig 6.6 Ammonia Level High

The display shows real-time environmental conditions with a temperature of 32.5°C and humidity of 38%. The message “AMMONIA HIGH” indicates that the ammonia level has exceeded the safe limit, which can be harmful to poultry. This alerts the system to take action, such as activating ventilation or neutralizing mechanisms, to restore safe air quality.

6.2 EXISTING SYSTEM VS PROPOSED SYSTEM

Parameter	Existing System	Proposed System
Energy Source	Manual / Basic Power	Smart IoT + Backup Supply
Control	Manual Monitoring	IoT Remote / Automated Control
Response Time	Periodic / Manual Checking	Real-time NH_3 Sensor Monitoring
Power Stability	Unstable	Stable Power Supply
Energy Utilization	Higher (Inefficient Usage)	90–95% Efficient (Optimized Consumption)
Operational Cost	High	30–50% Lower
Maintenance	Reactive	Predictive (IoT-based)

Table 6.1 Comparison Table

The table compares the existing system with the proposed automated ammonia detection and neutralizing system in poultry farms. In the existing system, energy is supplied through basic power sources and operations are carried out manually, whereas the proposed system uses smart IoT-based power management with backup supply. Monitoring and control in the existing system are manual, with periodic checking of ammonia levels, leading to delayed response and unstable air quality. In contrast, the proposed system uses real-time NH_3 sensors, enabling instant response and automatic control through ventilation and neutralizing mechanisms. The existing system has low accuracy, no data monitoring, and lacks alert systems, while the proposed system offers high-precision sensing, cloud-based monitoring (ThingSpeak), and alert notifications via buzzer or mobile**.**

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The proposed ammonia detection and neutralizing system provides an effective solution to maintain air quality in poultry farms. By continuously monitoring ammonia levels and automatically activating a neutralization mechanism, the system ensures a healthy environment for poultry birds. It reduces respiratory issues, decreases mortality rate, and improves overall productivity and profitability of the farm. Compared to traditional methods, the system offers real-time monitoring, automation, and better efficiency. Hence, it is a practical, cost-effective, and scalable solution for modern poultry farming.

7.2 FUTURE SCOPE

- Integration with IoT platforms for remote monitoring through mobile applications
- Addition of multiple gas sensors (H_2S, CO_2, PM) for complete air quality monitoring
- Implementation of AI/ML algorithms for predictive analysis and smart control
- Development of a solar-powered system for energy efficiency
- Design of a fully automatic ventilation + neutralization system
- Improvement in sensor accuracy using industrial-grade sensors
- Conversion into a commercial product for large-scale deployment

CHAPTER 8
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